

Name: Solution (please print)

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ECE 3364 – Quiz #3
Spring 2026

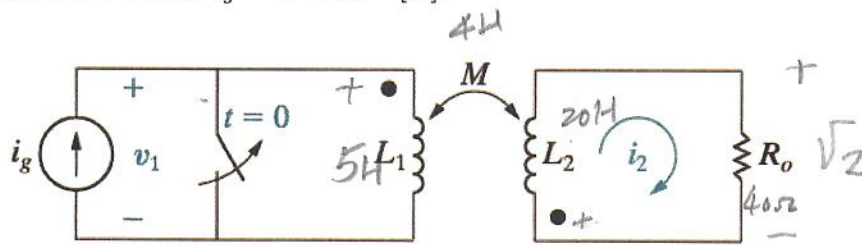
Do not open this quiz until you are told to begin.

1. Print your name, and sign your name, at the top of this page.
2. This quiz is closed book, closed notes. You may use one 8.5" × 11" crib sheet, or its equivalent. You may use a calculator. You should not use a cell phone, tablet computer, or laptop computer as you work on this quiz.
3. Show all work on these pages, and do NOT use backside of each page. Show all work necessary to complete the problem. A solution without the appropriate work shown will receive no credit. A solution which is not given in a reasonable order will lose credit. You may separate the pages as you work.
4. Show all units in solutions, intermediate results, and figures. Units in the quiz will be included between square brackets.
5. Do not use red ink. Do not use red pencil.
6. You will have 30 minutes to work on this quiz.

_____/100 (+10 bonus)

Room for extra work

There is no energy stored in the circuit shown in Figure 1 at the time the switch is opened ($t = 0$). The circuit parameters are $L_1 = 5$ [H], $L_2 = 20$ [H], $M = 4$ [H], and $R_o = 40$ [Ω]. The source current for $t \geq 0$ is $i_g = 8 - 8e^{-t}$ [A].



- a) (20 pts) Derive the differential equation that governs the behavior of i_2 .

KVL around the second loop.

$$\begin{cases} v_{L2} + i_2 R_o = 0 \\ v_{L2} = L_2 \frac{di_2}{dt} + M \frac{di_g}{dt} \end{cases}$$

eliminate v_{L2} :

$$i_2 R_o + L_2 \frac{di_2}{dt} + M \frac{di_g}{dt} = 0$$

$$40 i_2 + 20 \frac{di_2}{dt} + 4 \frac{di_g}{dt} = 0$$

$$\frac{di_2}{dt} + 2 i_2 = -0.2 \frac{di_g}{dt}$$

- b) (20 pts) Show that $i_2(t) = 1.6(e^{-t} - e^{-2t})$ [A] for $t \geq 0$ satisfies the differential equation you derived in part (a).

Need to verify $i_2(t) = 1.6(e^{-2t} - e^{-t})$ A satisfy (the original $i_2(t)$)

$$\begin{aligned} \text{LHS: } & 1.6((-2)e^{-2t} - (-1)e^{-t}) + 2 \cdot (1.6)(e^{-2t} - e^{-t}) \\ &= -3.2e^{-2t} + 1.6e^{-t} + 3.2e^{-2t} - 3.2e^{-t} \\ &= -1.6e^{-t} \end{aligned}$$

does NOT satisfy.

$$\text{RHS } -0.2 \cdot (0 - (-1)8e^{-t}) = -1.6e^{-t}$$

$$\therefore \text{LHS} = \text{RHS}$$

$$\text{Actual } i_2(t) = -1.6(e^{-t} - e^{-2t}) = 1.6(e^{-2t} - e^{-t})$$

c) Expression for $\bar{v}_1$

Note, Both current enter dotted terminal, so the mutual term in $\bar{v}_L$, adds!

$$\begin{aligned}\bar{v}_1 &= L_1 \frac{di_1}{dt} + M \frac{di_2}{dt} \\ &= 5 \cdot (0 + 8e^{-t}) + 4(1.6((-2)e^{-2t} - (-1)e^{-t})) \\ &= 40e^{-t} + 6.4(-2e^{-2t} + e^{-t})\end{aligned}$$

① $= 46.4e^{-t} - 12.8e^{-2t} \text{ [V]} \quad t \geq 0$

if you use $\bar{v}_2(t) = 1.6(e^{-t} - e^{-2t})$, then your alternative solution will be

$$\begin{aligned}\bar{v}_1 &= 5(0 + 8e^{-t}) + 4(1.6)(-e^{-t} + 2e^{-2t}) \\ &= 40e^{-t} + 12.8e^{-2t} - 6.4e^{-t}\end{aligned}$$

② $= 33.6e^{-t} + 12.8e^{-2t} \text{ [V]} \quad t \geq 0$

d) Initial value of $\bar{v}_1$.

This is follow up of part (c). Once you have the formula of $\bar{v}_1(t)$ function, $t=0^-$, then just plug in t

Case ①

$$\begin{aligned}\bar{v}_1(0) &= 46.4e^{-t} - 12.8e^{-2t} \Big|_{t=0} \\ &= 46.4 - 12.8 \\ &= 33.6 \text{ [V]}\end{aligned}$$

Case ②

$$\begin{aligned}\bar{v}_1(0) &= 33.6e^{-t} + 12.8e^{-2t} \Big|_{t=0} \\ &= 33.6 + 12.8 \\ &= 46.4 \text{ [V]}\end{aligned}$$

e) Instant energy in coupled coils

With both $\vec{v}_1 = i\vec{g}$ enter the dotted L_1
 $\vec{v}_2$ entered the dotted L_2

The mutual energy term M is positive

$$W(t) = \frac{1}{2} L_1 \vec{v}_1^2 + \frac{1}{2} L_2 \vec{v}_2^2 + M \vec{v}_1 \vec{v}_2$$

Case ①

$$\vec{v}_2(t) = 1.6(e^{-2t} - e^{-t})$$

plug in values

$$\begin{aligned} W(t) &= \frac{1}{2}(5)(8 - 8e^{-t})^2 \\ &\quad + \frac{1}{2}(20)(1.6(e^{-2t} - e^{-t}))^2 \\ &\quad + 4(8 - 8e^{-t})(1.6)(e^{-2t} - e^{-t}) \\ &= 160 - 371.2e^{-t} + 288e^{-2t} \\ &\quad - 102.4e^{-3t} + 25.6e^{-4t} \text{ (J)} \end{aligned}$$

$$\text{check: } W(0) = 0$$

$$W(\infty) = 160$$

Case ②

$$\vec{v}_2(t) = 1.6(e^{-t} - e^{-2t})$$

plug in value

$$\begin{aligned} W(t) &= \frac{1}{2}(5)(8 - 8e^{-t})^2 \\ &\quad + \frac{1}{2}(20)(1.6(e^{-t} - e^{-2t}))^2 \\ &\quad + 4(8 - 8e^{-t})(1.6)(e^{-t} - e^{-2t}) \\ &\quad \vdots \\ &= 160 - 268.8e^{-t} + 83.2e^{-2t} \\ &\quad + 25.6e^{-4t} \text{ (J)} \end{aligned}$$

$$W(0) = 0$$

$$W(\infty) = 160$$

f) steady state energy

as $t \rightarrow \infty$

$$\dot{v}_g = 8 - 8e^{-t} = 8, \quad \dot{v}_g \rightarrow 8$$

$\dot{v}_2 \rightarrow 0$ in both case ① and case ②

$$\therefore W(\infty) = \frac{1}{2} L_1 \dot{v}_g^2 + \frac{1}{2} L_2 \dot{v}_2^2 + M \dot{v}_g \dot{v}_2 \left| \begin{array}{l} \dot{v}_g \rightarrow 8 \\ \dot{v}_2 \rightarrow 0 \end{array} \right.$$

$$= \frac{1}{2} (5) \cdot 8^2 + 0 + 0$$

$$= 160 \text{ [J]}$$

This is consistent with part e)